

Chapter 25 — Molecular Genetics and Cancer Biology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **TP53**: commonest mutated gene in cancer; mutated in >50% of malignancies.
- **RB1 phosphorylation** by cyclin D-CDK4/6 + E-CDK2 → E2F free → S phase.
- **Two hits** (Knudson) — exceptions: dominant negative (TP53), haploinsufficiency (p27).
- **GSTP1 hypermethylation**: commonest somatic change in prostate cancer (>90%).
- **TMPRSS2-ERG**: ~half of prostate cancers; commonest fusion.
- **BRCA1/2 loss** → **PARP inhibitor** (approved mPC); ATM, CDK12 → little benefit.
- **MMR deficiency/MSI** → **immunotherapy** (tissue-agnostic).
- **VHL loss** → **HIF pseudohypoxia**; ccRCC 3p loss near-universal; belzutifan = HIF-2α inhibitor.
- **FGFR**: 15–20% advanced bladder UC, up to 37% UTUC → erdafitinib; test all advanced UC.
- **Predictive marker** needs biomarker–treatment interaction; prognostic = parallel lines.

BASIC MOLECULAR GENETICS

- **RNA polymerase II** makes pre-mRNA; cap, poly-A, splice.
- **AR splice variants** lack ligand-binding domain → castration resistance.
- **Start codon AUG** (methionine); three stop codons.
- **rRNA** ~80% of cellular RNA; tRNA 75–85 bases.
- **Initiation** rate-limiting; elongation fastest.

SOMATIC AND GERMLINE TESTING

- **Somatic**: tumour tissue → targeted therapy. **Germline**: blood/saliva → risk, family.

INACTIVATION OF TUMOUR SUPPRESSOR GENES AND ACTIVATION OF ONCOGENES

- **Point mutations**: silent, missense, nonsense (premature stop).
- **Indel** → frameshift.
- **Suppressor loss routes (4)**: homozygous deletion; LOH + mutation; two mutations; LOH + methylation.
- **Oncogene activation (3)**: mutation, amplification, rearrangement.
- **MET**: germline activating → hereditary papillary RCC.
- **Isochromosome**: one arm lost, other duplicated; **ring** = intrachromosomal fusion; **dicentric** = end-to-end.

CELL CYCLE DEREGLATION

- **Cycle** ~24 h; G0 = reversible quiescence (default).
- **Cyclin order**: D → E → A → B; 12 mammalian cyclins.
- **Cyclin B-CDK1** drives mitosis (lamin breakdown).
- **G1/S**: external signals + damage; **G2/M**: damage only.
- **S-phase arrest**: ATM → Cdc25A destroyed → CDK2 off.
- **G2/M**: ATM → CHEK1/2 → Cdc25 sequestered by 14-3-3σ.
- **TP53** mediates both G1/S and G2/M arrest.
- **Spindle checkpoint** blocks anaphase-promoting complex.
- **Aneuploidy**: near-universal in solid tumours.
- ↓ **MAD1/MAD2** in RCC; ↓ **MAD2** testis; hBUBR1 mutations rare.
- **Cip/Kip** (p21, p27, p57) broad; **INK4** (p15, p16, p18, p19) CDK4/6 only.
- **p21** ↑ after DNA damage; **p27** high in quiescence.
- **RB1** phosphorylated at eight sites; E2F + DP1–3.

DNA DAMAGE AND REPAIR

- **Replication error** ~3 bases per division; genome >3 × 10⁹ bases.
- **Oxidative lesions** ~10 000 per cell per day.
- **NER**: bulky/UV; **BER**: oxidative/deamination (OGG1, MUTYH).

- **MMR loss** → ~100-fold ↑ mutation rate; MLH1.
- **HR**: precise, sister template, S/G2 only; BRCA1/2, RAD51.
- **NHEJ**: error-prone; Ku70/80, DNA-PKcs, ligase IV/XRCC4.
- **γH2AX** spreads up to 30 Mbp from break.
- **Loeb**: mutator phenotype explains excess mutations.

APOPTOSIS AND CELL DEATH

- **Initiator caspases** 8, 9, 10; **executioner** 3, 6, 7; ≥13 total.
- **Intrinsic**: cytochrome c + Apaf-1 = apoptosome → caspase-9.
- **Extrinsic**: FasL-CD95 → FADD → caspase-8; FLIP inhibits.
- **Bax** transactivated by TP53; **Bcl-2** binds Bax.
- **Bid** (cut by caspase-8) links extrinsic → intrinsic.
- **Telomere loss** ~100 bp per division.
- **Telomerase** high in 85–90% of tumours; ALT via ATRX/DAXX.
- **Short telomeres** predict prostate cancer recurrence.
- **SA-β-gal** marks senescence; SASP double-edged.
- **ADT, enzalutamide** → therapy-induced senescence.
- **Autophagy**: mTOR/AMPK → ULK1/2 → Beclin-1; chloroquine inhibits.
- **CpG islands** ~40% of promoters; tumour loss 20–50% methylcytosine.
- **RASSF1A** methylated up to 97% bladder; p16 27–60% UC.
- **LINE-1 hypomethylation** bladder cancer.
- **DNMT1** maintenance; **DNMT3A/3B** de novo.
- **Activating** H3K4me3; **repressive** H3K9me3, H3K27me3 (EZH2).
- **EZH2 high** → PSA recurrence; ↑ in CRPC.
- **lncRNA** >200 nt (HOTAIR, MALAT1); **miRNA** 18–25 nt; **piRNA** 24–31 nt.
- ~98% genome transcribed as ncRNA; <2% protein-coding.
- **VHL inactivated** ~60% RCC (mutation ~50%, methylation ~10%).
- **Warburg** (Nobel 1931): aerobic glycolysis; CA9, MCT4, NHE1.
- **Vasculogenic mimicry**: tumour cells line vessels.

NEXT-GENERATION SEQUENCING TECHNIQUES

- **Bisulfite**: unmethylated C → U; methylated unchanged.
- **ATAC-seq** = open chromatin; **ChIP-seq** = protein–DNA binding.
- **scRNA-seq**: barcode cells before library.

ADVANCES IN THE UNDERSTANDING OF MOLECULAR GENETICS IN GENITOURINARY TUMOURS

- **AVPC**: loss of ≥2 of TP53, RB1, PTEN; overlaps NEPC.
- **SPOP**: commonest mutated suppressor, primary prostate cancer.
- **MYC** on 8q; 8q24 germline risk locus.
- **AR amplification** (incl. enhancer, ecDNA) → ADT resistance.

- **ccRCC** 75–80%; papillary 15–20%.
- **VHL** 60–70% mutated + 5–10% silenced in ccRCC.
- **PBRM1** ~40% (better); **BAP1**, **SETD2** ~10% (worse).
- **VHL syndrome** → >70% lifetime RCC.
- **Chromophobe**: distal nephron; mitochondrial DNA, mTOR.
- **Translocation RCC**: MiT/TFE; young women.
- **Sarcomatoid**: PD-L1+, immunotherapy-responsive.
- **UC site**: 90% bladder, 10% upper tract.
- **Consensus MIBC**: six subtypes; basal/squamous 35% largest; NE-like 3%.
- **HER2** 9–12% MIBC; no approved agent.
- **ERCC2** 10–18% → cisplatin sensitivity; ~10% germline DDR.
- **TGCT**: 12p gain; KIT/KRAS/NRAS; TP53 rare.
- **miR-371a-3p**: >90% sensitivity and specificity.
- **HPRC MET** → papillary type 1; **HLRCC FH** → type 2, collecting duct.
- **BHD (FLCN)** → hybrid oncocytic, chromophobe; **HP-JT HRPT2**.
- **HOXB13, BRCA2, Lynch** → prostate; Lynch → urothelial.

GENERAL PRINCIPLES OF CLINICAL NGS ASSAYS

- **VAF** = allele fraction × purity: $0.5 \times 0.6 = 30\%$.
- **VAF ~50%** → heterozygous germline; **~100%** → homozygous germline.
- **High VAF** = clonal; low = subclonal.
- **WES** cuts cost ~2 orders vs WGS.
- **CHIP** mimics tumour variants at single-digit VAF in cfDNA.
- **MRD**: WES primary → bespoke panel → ultra-deep ctDNA.
- **AR-V7** on CTCs; EpCAM enrichment.

CLINICAL APPLICATION OF MOLECULAR BIOMARKERS IN GENITOURINARY CANCERS

- **Short-term ADT**: unfavourable intermediate risk only; long-term: high risk.
- **ARR** = baseline risk × RRR: $30\% \times 50\% = 15\%$; $3\% \rightarrow 1.5\%$.
- **PORTOS** predicts PORT dose-escalation benefit (SAKK 08/09).
- **HSD3B1 1245A>C** → ADT-resistant, abiraterone-sensitive.

CLASSIC TRAPS

- **Trap**: every suppressor needs two hits — **Actually**: TP53 dominant negative; p27 haploinsufficient.
- **Trap**: G2/M senses growth factors — **Actually**: DNA damage only.
- **Trap**: NHEJ is precise — **Actually**: error-prone; HR precise (S/G2).
- **Trap**: extrinsic apoptosis needs TP53 — **Actually**: TP53-independent.
- **Trap**: ATM loss = BRCA for PARP — **Actually**: little benefit.
- **Trap**: TP53 common in testis — **Actually**: rare.
- **Trap**: PBRM1 = poor prognosis — **Actually**: better; BAP1/SETD2 worse.
- **Trap**: siRNA broad — **Actually**: one target; miRNA many.
- **Trap**: VAF 50% = somatic clonal — **Actually**: suspect germline without matched normal.
- **Trap**: prognostic = predictive — **Actually**: needs treatment interaction.